22. Ropes, knots and lashings — essentials for shelter, loads, rescue and fishing

Oracle OS guide. Based on US Army FM 3-97.61 (Military Mountaineering), the FAO Fisherman's Workbook and A. Hyatt Verrill's "Knots, Splices and Rope Work" (in the library **with step-by-step illustrations**; also the "Knots and ropes (zimgit)" collection in the encyclopedia reader). To hang people on a rope (climbing, lowering, rescue at height) use certified gear and training: one mistake kills.

Know your rope

- **Synthetic** (nylon, polyester, polypropylene): strong, rot-proof. Nylon stretches (absorbs shock); polyester stretches little (lashings); polypropylene floats (water rescue) but degrades in sunlight.
- **Natural** (sisal, hemp, cotton): weaker, rot when damp, but hold knots well.
- **Static** (low stretch) for rescue and loads; **dynamic** (stretchy) only for lead climbing.
- A knot cuts rope strength by **20–50%**. Load only a fraction of breaking strength: **1/5 for loads**, far less for people.
- Care: keep away from sand, chemicals, battery acid, sharp edges (pad them), prolonged sun. Dry in shade. Retire rope with stiff, soft, fuzzy or melted sections.
- Ends: melt synthetics or **whip** them with thread so they don't fray.

Terms

Working end, **standing part**, **bight** (U-shaped fold), **loop** (crossed). Always **dress** (tidy and tighten) the knot and leave a tail of at least 10 rope diameters.

The most useful knots

1. **Figure-eight stopper**: make a bight, take the working end round the standing part and back through the bight — it forms an "8". Stops the rope pulling through.
2. **Figure-eight follow-through** (strong fixed loop): tie a loose figure-eight 1 m from the end, pass the end around the anchor (or harness) and **retrace the eight exactly**. Standard tie-in knot; easy to check and untie.
3. **Bowline** (non-slip loop): small loop in the standing part ("the hole"), the end ("the rabbit") comes up through the hole, round the tree (standing part) and back down the hole. Dress it. Can loosen without load: back it up with a stopper.
4. **Clove hitch**: two overlapping turns round a post, the second tucked under the first. Quick start for lashings and tying to posts; adjustable.
5. **Round turn and two half hitches**: two full turns round the post, then two half hitches on the standing part. Holds under load and releases easily. Great for boats and tarps.
6. **Taut-line hitch**: round the peg, two turns on the standing part inside the loop and one outside. Slides to tension a guy line and grips under load.
7. **Trucker's hitch**: make a fixed loop in the standing part, pass the end round the hook/peg, back through the loop and pull — **3:1 advantage**. Finish with two half hitches. For loads, clotheslines and shelter ridgelines.

8. **Sheet bend**: joins ropes of different thickness. Bight in the thick rope; thin rope comes up through the bight, round both legs and under itself. **Double sheet bend** for more security.
9. **Double fisherman's**: joins two equal ropes under load (closes cord loops). Each end ties a double overhand (two turns) round the other; they jam together.
10. **Prusik**: loop of thinner cord (5–7 mm on 10–11 mm rope) wraps the main rope 3 times and passes through itself. Grips when loaded, slides when not: ascending, backing up a lower, rescue systems.
11. **Munter hitch**: on a carabiner, creates friction to hold or lower a load.
12. **Timber hitch**: round a log, the end wraps round itself 3 times. For dragging or lifting logs; add a half hitch ahead to steer.
13. **Reef knot**: right over left, then left over right. **Bandages and parcels only** — slips under load, never to join load-bearing ropes.
14. **Improved clinch** (fishing): line through the hook eye, 5–7 turns round the standing line, back through the first loop by the eye, then through the big loop; wet and pull tight.

Lashings for building (shelters, tripods, stretchers, rafts)

- **Square lashing** (crossed poles): clove hitch on one pole, 3–4 wraps around both, 2–3 **frapping** turns between the poles, finish with a clove hitch.
- **Diagonal lashing**: when poles want to spring apart — diagonal wraps one way, then the other, frapping, clove hitch.
- **Shear / tripod lashing**: poles side by side, 6–8 loose wraps, 2 frapping turns between them; open into an A-frame or tripod (hang a pot, support a shelter, make a trestle).
- **Shelter ridgeline**: rope between two trees with a round turn at one end and a trucker's hitch at the other; tarp over, corners tied with taut-line hitches.

Mechanical advantage (moving weight)

- A fixed pulley only changes direction. A **Z-drag (3:1)**: rope tied to the load runs through a pulley/carabiner at the anchor, back to a second pulley clipped with a prusik onto the rope near the load, then to the pullers. One third of the effort (minus friction).
- Anchors: thick living trees, large boulders, vehicles (braked). Share the load across more than one anchor.

Making cordage without rope

- Fibres: dried nettles, inner bark of lime/willow/oak, cattail leaves, palm raffia, plastic bags cut into strips, cloth strips.
- **Reverse wrap**: fold a bundle of fibres in half; twist the top strand away from you and cross it over the other toward you; repeat alternately. The cord locks itself. Splice in new fibres before a strand runs out.
- A three-strand braid works too but is weaker.

Practise

Practise each knot until you can tie it eyes closed and with gloves. Priority: figure-eight follow-through, bowline, round turn and two half hitches, trucker's hitch, sheet bend, prusik.